

Demand for multi-scale weather data for regional crop modeling



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ABSTRACT

A spatial resolution needs to be determined prior to using models to simulate crop yields at a regional scale, but a dilemma exists in compromising between different demands. A fine spatial resolution demands extensive computation load for input data assembly, model runs, and output analysis. A coarse spatial resolution could result in loss of spatial detail in variability. This paper studied the impact of spatial resolution, data aggregation and spatial heterogeneity of weather data on simulations of crop yields, thus providing guidelines for choosing a proper spatial resolution for simulations of crop yields at regional scale. Using a process-based crop model SIMPLACE (LINTUL2) and daily weather data at 1 km resolution we simulated a continuous rainfed winter wheat cropping system at the national scale of Germany. Then we aggregated the weather data to four resolutions from 10 to 100 km, repeated the simulation, compared them with the 1 km results, and correlated the difference with the intra-pixel heterogeneity quantified by an ensemble of four semivariogram models. Aggregation of weather data had small effects over regions with a flat terrain located in northern Germany, but large effects over southern regions with a complex topography. The spatial distribution of yield bias at different spatial resolutions was consistent with the intra-pixel spatial heterogeneity of the terrain and a log–log linear relationship between them was established. By using this relationship we demonstrated the way to optimize the model resolution to minimize both the number of simulation runs and the expected loss of spatial detail in variability due to aggregation effects. We concluded that a high spatial resolution is desired for regions with high spatial environmental heterogeneity, and vice versa. This calls for the development of multi-scale approaches in regional and global crop modeling. The obtained results require substantiation for other production situations, crops, output variables and for different crop models.

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1. Introduction

Crop models are becoming essential tools to study the interactions between plant, climate, soil and management practices in agricultural systems at different scales (Fischer et al., 2005; van Ittersum et al., 2008) to support policy and decision making for food security and land-use related management (De Vries et al., 1997; Rabbinge and Van Diepen, 2000). To apply crop models at regional scale, a level of homogeneity of spatial units or resolution needs to be assumed (Harrison et al., 2000), since the spatial heterogeneity of environmental conditions is ubiquitous (Wu, 2004). Once the resolution is determined, the input data such as climate and soil conditions, and management practices need to be aggregated or disaggregated to the same resolution to drive the crop

models (Priya and Shibasaki, 2001). At a certain spatial extent, the resolution determines the detail in spatial variability that can be captured as well as the computational load. A fine resolution could capture more spatial detail, but it raises the demand for data storage, computational costs and efforts for results analysis (Zhao et al., 2013a). A coarse resolution requires less computing resources, but loses information in spatial variability.

The layers of coarse resolution are inherent or can be produced from layers of fine resolution by spatial aggregation (van Bussel et al., 2011). Spatial aggregation could be conducted by different methods such as spatial mean, median, and majority (Bian and Butler, 1999). No matter which method is adopted, loss of the spatial variability is inevitable (Li and Reynolds, 1995), because the resultant data uses one value to represent many others (Bian and Butler, 1999; Orcutt et al., 1968). The magnitude of the spatial variability loss varies with the features of the original data and the resolution differences. If the resultant area (also referred to as pixel) is homogeneous or covers a range of smaller pixels with similar

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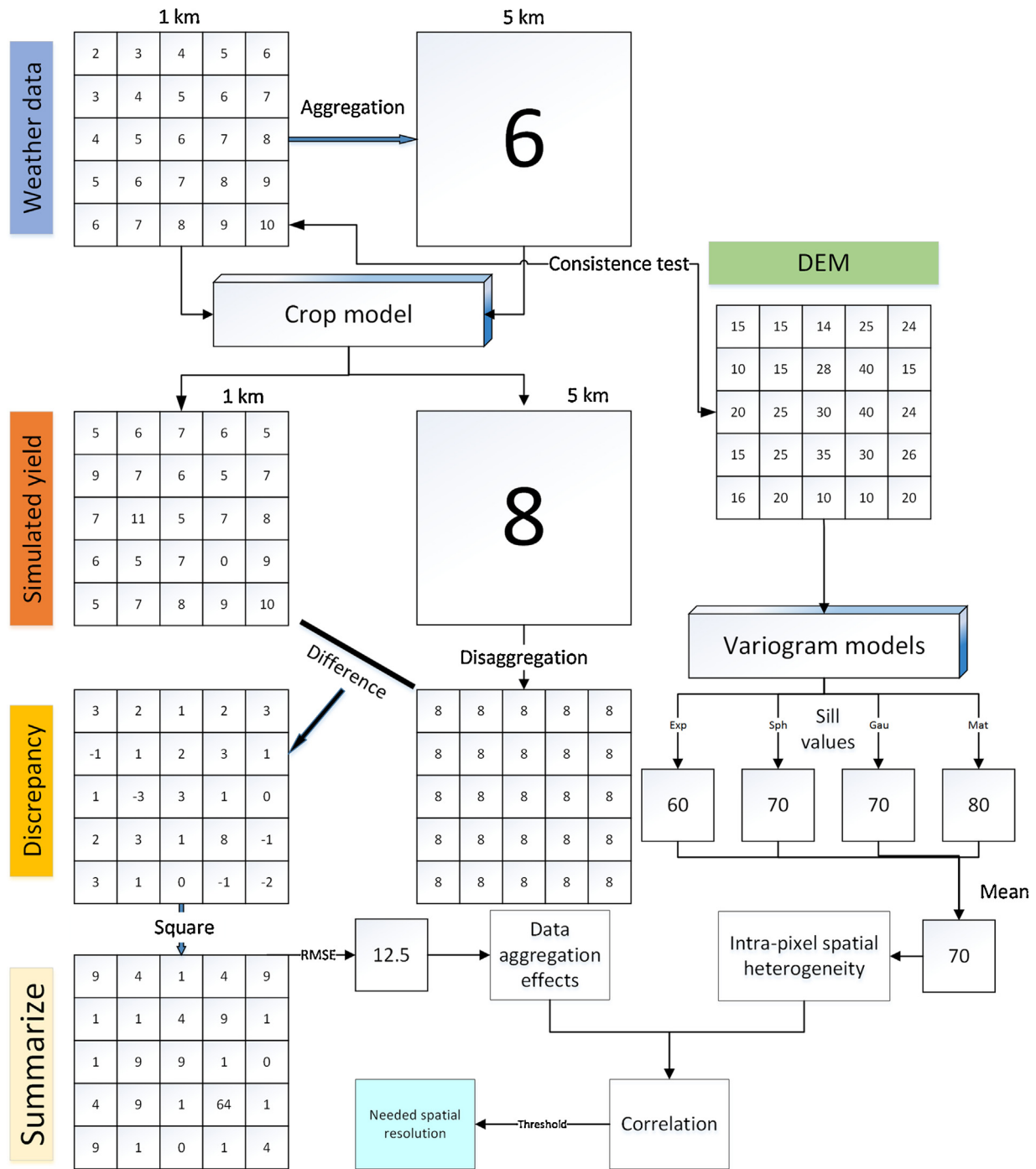


Fig. 1. Workflow of analysis. It schematically represents the data aggregation, disaggregation of the simulated results, bias calculation, and spatial heterogeneity quantification. Exp, Sph, Gau, and Mat refer to Exponential, Spherical, Gaussian, and Matérn models, which are used to fit the semivariogram. Note that a grid cell size of 5 km is only used for illustration purpose and not used in the present analysis.

values, the information loss would be minor. In contrast, the loss would be enormous if the covered smaller pixels are extremely heterogeneous. Here we term the variability and structure of the smaller pixels covered by the coarser resolution pixel as intra-pixel spatial heterogeneity. The loss of the intra-pixel spatial heterogeneity could be propagated by the crop model, thus affecting the simulated results such as crop yields (Guisan et al., 2007).

The effects of spatial resolution of weather data on simulated crop yields or crop phenology have been extensively studied (Angulo et al., 2013b; Folberth et al., 2012; Mearns et al., 1999; Mummery and Battaglia, 2002; van Bussel et al., 2011). These

studies evaluated the effects by comparing the results simulated by input data at different spatial resolutions and found relatively small effects of weather data aggregation (Hansen and Jones, 2000; Nendel et al., 2013). Aggregation effects on mean values or medians are diminished when the response of the crop model to the change of weather data is close to linear or the non-linear response is not significant. This could result in a conclusion that minor aggregation effects are caused by input data aggregation.

The aim of spatially explicit modeling is capturing the spatial heterogeneity, which should be a priority in determining the spatial resolution (Di Vittorio and Miller, 2014). However, to our

best knowledge no method has been developed to guide the choice of spatial resolution for regional scale simulation, because quantification of spatial heterogeneity of long-term time series of daily weather data is difficult. Weather data consisting of a set of variables, e.g. temperature, radiation, precipitation, and a high temporal frequency, often daily, is required for process-based crop models (Asseng et al., 2013; Balkovič et al., 2013). Within the spatial extent of a pixel (e.g. 10 km), the long-term weather condition is greatly influenced by the local topographic features (Ashcroft and Gollan, 2012). Topographic conditions are frequently used to correct the layers of weather data (Hancock and Hutchinson, 2006), thus long-term weather conditions are highly correlated with terrain features. Using the spatial heterogeneity of the topography to surrogate the weather conditions has the potential to bypass the difficulties in quantifying the spatial heterogeneity of different layers of weather variables (Kerr, 1993; Seoane et al., 2004). However, this has not been tested yet to inform about the required spatial resolution for simulating crop yield at regional scale using crop models. This paper aims to improve the understanding of the required spatial resolution of weather data for the simulation of regional crop yields and how the proper resolution can be determined.

2. Methods

2.1. General workflow of analysis

The general workflow of the simulation analysis is presented in Fig. 1. We produced 31-year weather data at 1 km resolution for the whole Germany and aggregated weather data into resolutions of 10, 25, 50, and 100 km. Using these data, we simulated a 30-year continuous cropping system of winter wheat, with a process-based crop model LINTUL2 (Light INTERception and Utilization) (van Ittersum et al., 2003; van Oijen and Lefelaar, 2008) under rainfed condition. Three regions, respectively, representing homogeneous, moderately heterogeneous, and notably heterogeneous conditions of terrain were selected to study the possible relationship between terrain conditions and weather conditions, and aggregation effects on the temporal variation of simulated yields. Semivariogram models were then used to quantify intra-pixel spatial heterogeneity of terrain at different aggregation levels (Fig. 1). Impacts of input data aggregation on the simulated crop yields were quantified by calculating the spatial bias of crop yields simulated with high resolution and aggregated weather data. We correlated the intra-pixel spatial heterogeneity of terrain with the yield difference and demonstrated how this relationship can be used to select the spatial resolution required to constrain the intra-pixel information loss to a specified threshold.

2.2. Study area, topography and climate

The study was performed for Germany with a territory area of 357,021 km², including 349,223 km² of land and 7798 km² of water bodies. The northern part of Germany is located in the North European Plain and has a flat terrain with altitudes between 0 and 150 m above the sea level (Fig. 2). The middle-central part is characterized by landforms of intermittent mountains while the flat topography from the north is stretching to the east and west of central Germany. The southern region's landforms are characterized by many hills and mountain ranges. The Alps Mountains cross the southern border to Austria and Switzerland and form the highest terrain of Germany. In Germany, the temperature can become very high in the summer, but winter is normally mild cold because of the influence from the North Atlantic Current. With the effects of the countervailing continental climate, during winter time the southern

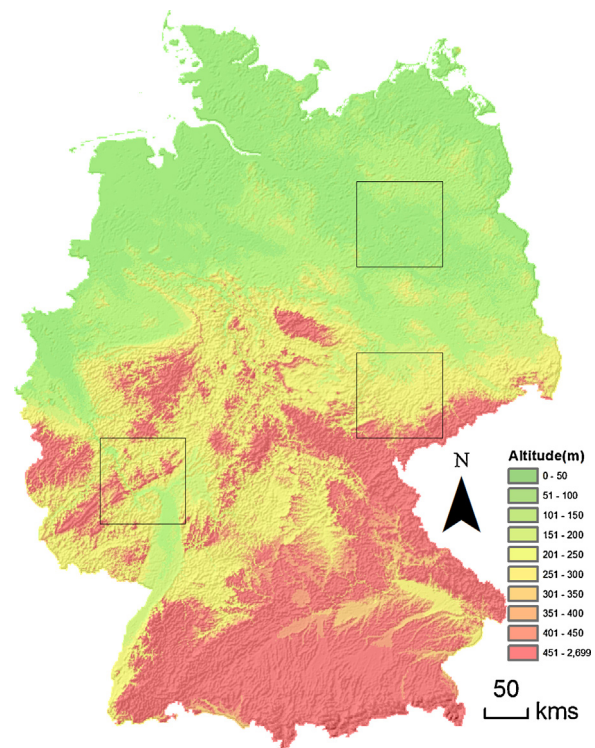


Fig. 2. The digital elevation model (DEM) of the study area, Germany (data source: Federal Agency for Cartography and Geodesy, Germany, <http://www.bkg.bund.de>). The boxes in the map show the locations of three 100 km × 100 km regions to represent conditions of homogeneous, moderately heterogeneous, and notably heterogeneous (from top to bottom) used for further analysis in this study.

mountainous areas experience prolonged cold, with temperatures around the freezing point, though hot days can also happen in this area during the summer time. The mean annual precipitation sum shows a high spatial variation. In the Alpine regions annual precipitation sum is very high and frequently exceeds 2000 mm, while annual precipitation sum in the driest regions in the North German Plain is between 500 mm and 750 mm.

2.3. Crop model, calibration and evaluation

The process-based crop model of LINTUL2 was employed in this study. LINTUL2 is a generic crop model that can simulate potential (Spitters, 1989) and water limited crop yields (Gaiser et al., 2013; van Oijen and Lefelaar, 2008). It was derived from LINTUL1 by adding a water balance model (Spitters, 1987). Ample nutrient supply and optimal crop management are assumed in the model therefore the crop does not suffer from nutrient scarcity, weed competition, diseases and pests. LINTUL2 simulates crop phenology, photosynthesis, dry matter production and partitioning, evapotranspiration and soil water balance at a daily time step. The phenology is simulated by the accumulation of thermal time above a defined base temperature. The thermal time is then modified by the sensitivity to vernalization and photoperiod. Photosynthesis and total crop growth rate are calculated by multiplying the intercepted light and light use efficiency (LUE). Total crop growth, root-shoot partitioning and leaf area expansion are further influenced by water stress. Water stress occurs when the available soil water is between a defined critical point and wilting point or higher than field capacity (logging). The critical point is a crop specific value depending on the drought tolerance of the crop. In the soil water balance model the potential transpiration is calculated via the Penman equation (Penman, 1963). The simple tipping bucket model is used to simulate the soil water dynamics and the amount

Table 1
Genetic and soil parameters of LINTUL2 used in this study.

Name	Description	Unit	Value
Crop parameters			
Thermal time anthesis	Temperature sum from emergence to anthesis	°C d ⁻¹	470
Thermal time maturity	Temperature sum from anthesis to maturity	°C d ⁻¹	402
Drought tolerance	Adjust of drought stress	–	1
Base temp before ant	Base temperature before anthesis	°C	1
Base temp after ant	Base temperature after anthesis	°C	9
LUE	Light use efficiency	g MJ ⁻¹	2.8
SLA	Specific leaf area	m ² g ⁻¹	0.022
Lai critical	Critical leaf area beyond which leaves die due to self-shading	m ² m ⁻²	4
LAI	Initial Leaf Area Index	m ² m ⁻²	0.025
TRANCO	Transpiration constant indicating the level of drought tolerance	mm	5.0
Vernal days max	Maximum days for vernalisation	day	30
Vernalization T _{low}	Vernalisation temperature lower limit	°C	4
Vernalization T _{high}	Vernalisation temperature upper limit	°C	10
ROOTDI	Initial rooting depth	m	0.1
ROOTDM	Maximum rooting depth	m	1.2
RRDMAX	Maximum rate of increase in rooting depth	m	0.005
Soil parameters			
INISW	Initial soil water content	m ³ m ⁻³	0.23
AD	Water content of air dry soil	m ³ m ⁻³	0.05
WP	Water content at wilting point	m ³ m ⁻³	0.07
FC	Field capacity	m ³ m ⁻³	0.28
WET	Critical soil water content for transpiration reduction due to waterlogging	m ³ m ⁻³	0.34
ST	Water content at full saturation	m ³ m ⁻³	0.38

of water available to the crop during the growing season (Keulen, 1975). LINTUL2 uses a single soil layer characterized by the parameters water content of air dry soil AD (m³ m⁻³), water content at wilting point WP (m³ m⁻³), at field capacity FC (m³ m⁻³), critical soil water content for transpiration reduction due to waterlogging WET (m³ m⁻³) and water content at full saturation ST (m³ m⁻³) to mimic soil water flows (Table 1). In this study, the soil water was reset to the initial soil water content at sowing, thus the soil water balance in the fallow period was not simulated and did not affect the crop growth of the following season. LINTUL2 has been validated and widely applied in simulating crop growth under various environmental conditions and at different scales (Adam et al., 2011; Asseng et al., 2013; Ewert et al., 1999; Farré et al., 2000; Hijmans et al., 2003; Palosuo et al., 2011; Wolf et al., 2002).

The original LINTUL2 model was developed for single season simulations. To simulate a continuous cropping system, the model was embedded into a general modeling framework, SIMPLACE (Scientific Impact assessment and Modeling Platform for Advanced Crop and Ecosystem management) (Gaiser et al., 2013). SIMPLACE follows a modular approach and accommodates many models for comparison and integrated assessment of agroecosystems. It draws

on earlier experiences of component-based modeling of crop, cropping and ecosystems (Adam et al., 2012).

In this study, a single set of cultivar related parameters was calibrated against field data from three sites in the federal state of Saxony (Table 1). Using a trial-and-error strategy, the parameters were adjusted so that simulated results for phenology, biomass and yield could match observations for the period from 1999 to 2011 (LfULG, 2000–2012). The field experiments were conducted under rainfed conditions with ample nutrient supply and control of pests-diseases.

We evaluated the model performance by comparing to observed yields for 404 German districts for the period 1999–2011 (Statistische Ämter des Bundes und der Länder, 2013). The technology trend of the observed yields was removed by detrending. We masked out the grid cells of non-cropland area of the simulated 1 km yield (Fig. 4a) and aggregated the simulated yields (only cropland) for each district. Then we compared the spatial patterns of the observed and simulated yields (Fig. 4b and c). To test the temporal performance of the model for each county, we conducted a linear regression and Spearman's correlation test between the observed yield and the simulated yield in the same period, 1999–2011. Finally, we aggregated the simulated and observed yield for the whole study area and compared the overall temporal performance of the model. All yield aggregations were area weighted.

2.4. Weather, soil and management data

LINTUL2 was forced by daily weather data of maximum and minimum temperature, solar radiation, precipitation, and wind speed. Since we ran the model at different resolutions (i.e. 1, 10, 25, 50 and 100 km), gridded weather data were created by fusion of data from a range of sources. We first created the weather data at 1 km resolution and then aggregated them to coarse resolutions. The data sources used to create the 1 km gridded data included daily precipitation at 1 km resolution from REGNIE (REGionalisierung von NIEderschlagsdaten) (DWD, 2012), monthly gridded data at 1 km resolution of minimum and maximum temperature and sunshine hours, daily data from weather stations across Germany (Siebert and Ewert, 2012), and incoming solar radiation derived from satellite imagery (Posselt et al., 2011) provided by CM SAF (<http://wui.cmsaf.eu/safira/action/viewHome>, accessed 05/0.7/2012). More information about data sources, the procedures to generate the data and maps showing examples of the data can be found in the Supplement S1.

A single set of crop and soil parameter and sowing date was used for all simulations to focus on the effects from the change of weather data resolution (Mearns et al., 2001; Olesen et al., 2000). Crop parameters referred to a typical winter wheat cultivar grown in Germany (Table 1). The values of the soil parameters were sourced from the dominant soil type in the Germany soil database (Engelen and Wen, 1993; Table 1), while sowing date was set as the 274th day of each year.

2.5. Evaluation of the aggregation effects

In this study, the term aggregation referred to estimating weather data at low resolutions (WL, 10 km to 100 km) from weather data at high resolution (WH, 1 km). The pixel value of WL was computed as the mean of the pixels of WH over the same spatial extent (Fig. 1). For example, a pixel of 10 km (WL) corresponded to 100 pixels of 1 km (WH). The aggregation can be described by:

$$WL_d = n^{-1} \sum_{i=1}^n WH_{i,d} \quad (1)$$

where i denoted the pixel of the WH , n denoted the number of the pixels of WH , d denoted that the temporal resolution of the weather data was daily. The value of WH was assumed as “true”. The resultant difference between WH and WL was described by the root mean squared error ($RMSE$)

$$RMSE_d = \left(n^{-1} \sum_{i=1}^n (WH_{i,d} - WL_d)^2 \right)^{1/2} \quad (2)$$

To quantify the aggregation effects ($RMSE_d$) on the simulated yields, the daily WH and WL were used as the input data to simulate crop growth and yield:

$$YL_{yr} = f(WL_d) \quad (3)$$

$$YH_{i,yr} = f(WH_{i,d}) \quad (4)$$

where f denoted the crop model. The temporal resolution for the simulated yield YL and YH was yearly, indicated by the subscript yr . The aggregation effects on the simulated yield (yield bias due to aggregation) was described as:

$$\overline{\Delta Y_i} = \overline{YH_i} - \overline{YL} \quad (5)$$

where $\overline{YH_i} = m^{-1} \sum_{yr=1}^m (\overline{YH_{i,yr}})$, $\overline{YL} = m^{-1} \sum_{yr=1}^m (\overline{YL_{yr}})$, m was the number of years. In this study m was equal to 30, from 1982 to 2011. To calculate $\overline{\Delta Y_i}$, we disaggregated $\overline{YL}$ to the same resolution of $\overline{YH}$. The $\overline{\Delta Y_i}$ was named as intra-pixel bias of yield. Across the country boundary, the area of $\overline{YL}$ was larger than the areas of $\overline{YH}$. We discarded the values outside the $\overline{YH}$ boundary.

Since $\overline{\Delta Y_i}$ had both negative and positive values, at the scale of a pixel of coarse resolution (intra-pixel), the intra-pixel $RMSE$ ($RMSE_{intra}$) of the yield difference was used to indicate the aggregation effects on the simulated yields.

$$\overline{RMSE}_{intra} = \left(n^{-1} \sum_{i=1}^n \overline{\Delta Y_i}^2 \right)^{1/2} \quad (6)$$

where i denoted the pixels of the WH , n denoted the number of the pixels of WH inside WL .

2.6. Intra-pixel spatial heterogeneity and semivariogram model

The data aggregation replaced multiple pixels of WH by one of WL . Each pixel WL was the average of multiple pixels of WH . Thus, the $RMSE$ between WH and WL was proportional to the intra-pixel spatial heterogeneity of the WH pixels. In this study, the intra-pixel spatial heterogeneity was quantified by semivariogram models, using the pixel values with the 1 km resolution covered by the coarse resolution pixel (Fig. 3) (Curran, 1988; Pebesma, 2004). Two steps were involved in the modeling of the semivariogram: computing the empirical semivariogram and fitting the empirical semivariogram to estimate the *sill* value, which represented the spatial heterogeneity (Fig. 1). The empirical semivariogram quantified the mean of squared differences between values $z(u)$ and $z(u+h)$ of paired grids (x_u, x_{u+h}) separated by distance h . The empirical semivariogram was defined as:

$$\gamma(h) = \frac{1}{2} E(|z(u) - z(u+h)|^2) = \frac{1}{2N} \sum (z(u) - z(u+h))^2 \quad (7)$$

where u and N represented the location of source pixels and the number of the paired pixels, respectively. The location of the pixel was represented by the central coordinate of the pixels in a projected coordinate system of DHDN/3-degree Gauss–Kruger zone 3 (<http://spatialreference.org/ref/epsg/31467/>). The spatial distribution of the grid values was assumed isotropic and no direction for the semivariogram was specified. The maximum distance to compute the semivariogram was set to half of the pixel extent. For

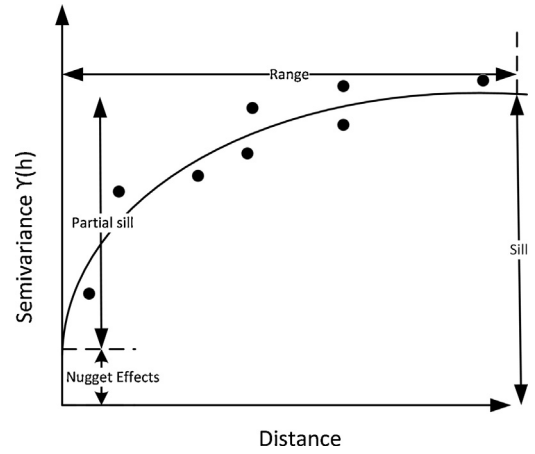


Fig. 3. Schematic representation of the semivariogram model. The dots represent the empirical semivariogram for a specific distance and the line represents the fitted semivariogram model that was used to quantify the *sill* value. The *sill* value represents the spatial heterogeneity.

example, 50 km was set as the maximum distance for the pixel of 100 km. The second step, fitting the empirical semivariogram for *sill* value calculation, was conducted after the empirical semivariogram was calculated. There are many models (e.g. Exponential, Spherical, Gaussian and Linear) that are able to fit the empirical semivariogram (Pebesma, 2004). Each of them has their merits and drawbacks. We individually tested these models and found in some cases considerable differences between these models. We therefore used the ensemble mean of the four frequently used models, i.e. Exponential, Spherical, Gaussian, and Matérn model, to obtain a better fit of the empirical semivariogram and robust quantifications of the *sill* (Pebesma, 2004; Stein, 1999). The mean of the *sill* values was used to represent the spatial heterogeneity.

To test the performance of the semivariogram model, we visually chose three regions with a spatial extent of 10,000 km² under homogenous, moderately heterogeneous and notably heterogeneous conditions. The empirical semivariogram was modeled for the terrain and for the weather variables of mean daily maximum temperature, mean daily minimum temperature, and annual precipitation for which we had data with a high spatial resolution available. We compared the empirical semivariogram among the terrain and weather factors to validate our hypothesis that at a certain extent the spatial heterogeneity of weather could be represented by the heterogeneity of elevation. If they were consistent, we used the intra-pixel spatial heterogeneity of the elevation to represent the heterogeneity of the weather variables. We fitted the *sill* values for all the pixels of the terrain using altitude values from digital elevation model (DEM) at resolutions of 10, 25, 50 and 100 km, corresponding to the resolution of aggregated weather data.

After the intra-pixel heterogeneity of terrain was quantified, it was correlated with the $RMSE$ of yield difference ($RMSE_{intra}$), pixel by pixel, resolution by resolution. We used the log function to consolidate the orders of magnitude of the intra-pixel spatial heterogeneity and the simulated yield difference.

2.7. Selecting model resolution according to the spatial heterogeneity of input data

To demonstrate how the model resolution could be selected to account for the local heterogeneity of the study areas, we calculated the intra-pixel $RMSE$ of yield difference ($RMSE_{intra}$) for each resolution of the weather input data. Then we defined two thresholds for $RMSE_{intra}$ (200 and 400 kg ha⁻¹ yr⁻¹) and computed the corresponding *sill* thresholds of the terrain by applying the regressed

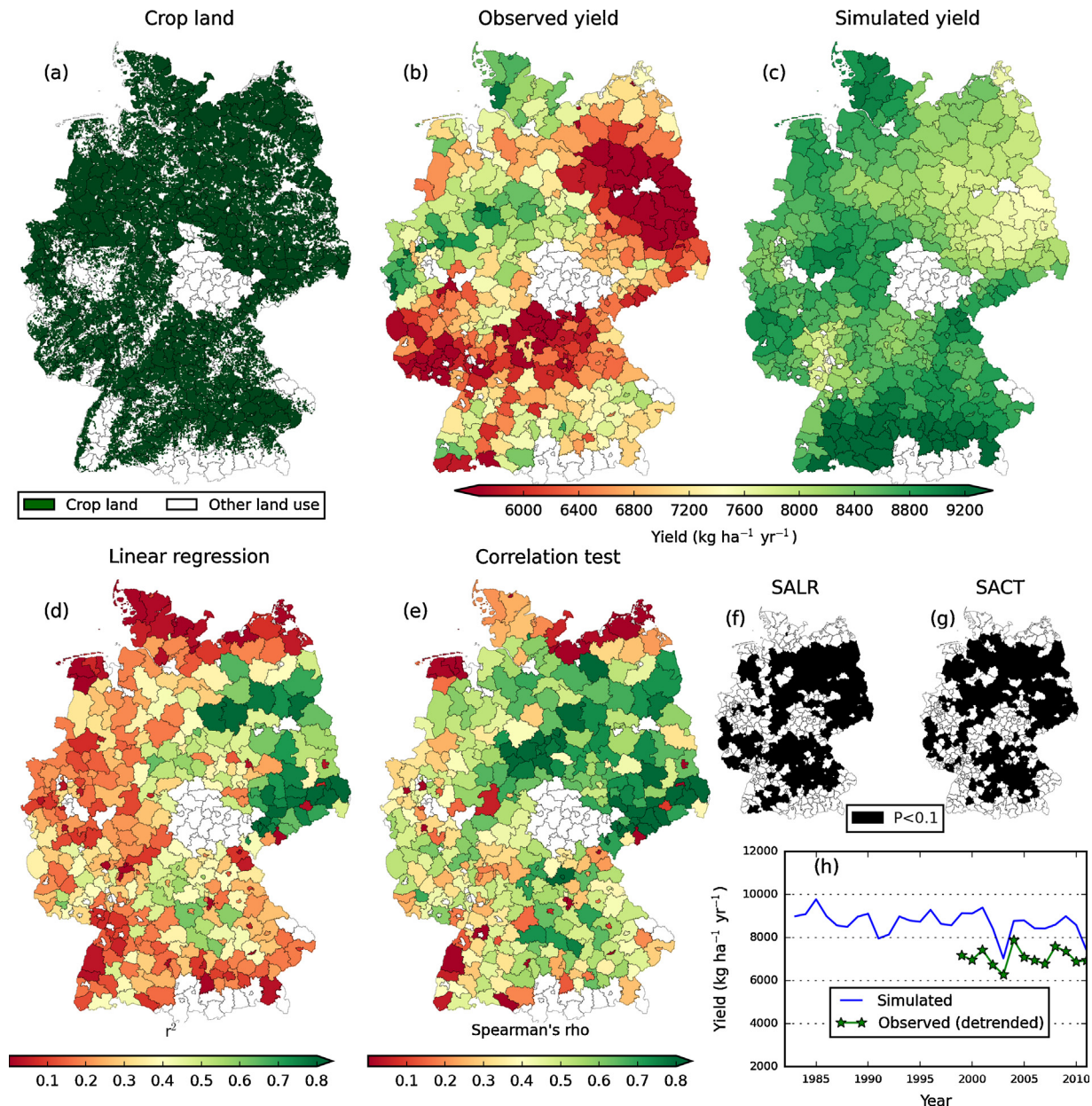


Fig. 4. Evaluation of the model performance by comparing spatial patterns of observed (b) and simulated yield (c), regression of simulated and observed yield (d and f), correlation test (e and g) and temporal variability (h). The maps show spatial distribution of crop land (a), average of observed yield over 1999–2011 (b), average of simulated yield over 1999–2011 (c), the goodness-of-fit of linear regression between observed and simulated yield (d), the Spearman's rank correlation coefficient between observed and simulated yield (e), the significant area of linear regression (SALR) and the significant area of correlation test (SACT). Note that zones with more than 15% agricultural land use were used for the evaluation, otherwise were left empty.

equations between intra-pixel heterogeneity of terrain (*sill*) and $\overline{RMSE}_{intra}$. Using the thresholds of the *sill*, we determined the spatial resolution of the weather data required to constrain $\overline{RMSE}_{intra}$ to the defined thresholds. To validate the resulting multi-resolution pattern computed from the *sill* thresholds we compared them with patterns calculated directly from the “true” $\overline{RMSE}_{intra}$ calculated by application of equation 6.

3. Results

3.1. Model evaluation

The model overestimated the yields for the majority of the study area, especially in southern regions (Fig. 4b and c). The spatial patterns between observed and simulated yield were similar.

Both the linear regression (r^2) and correlation test (ρ) showed that the model reproduced the temporal evolution quite well in the north-eastern regions, but relatively poor in the northern and south-western regions (Fig. 4d–g). The temporal variation of simulated yield was consistent with the historical observed yield but higher (Fig. 4h) which was expected as simulated yields refer to water-limited yields which are typically higher than actual observed yields under rainfed conditions.

For the three selected regions, the mean yield for a specific year was quite close across different resolutions, while the magnitudes (difference between maximum and minimum) of variation decreased with the coarsening of the resolution (Fig. 5). For a specific year, the magnitude of variation increased, when changing from regions with homogeneous condition (Fig. 5a) to notably heterogeneous (Fig. 5c).

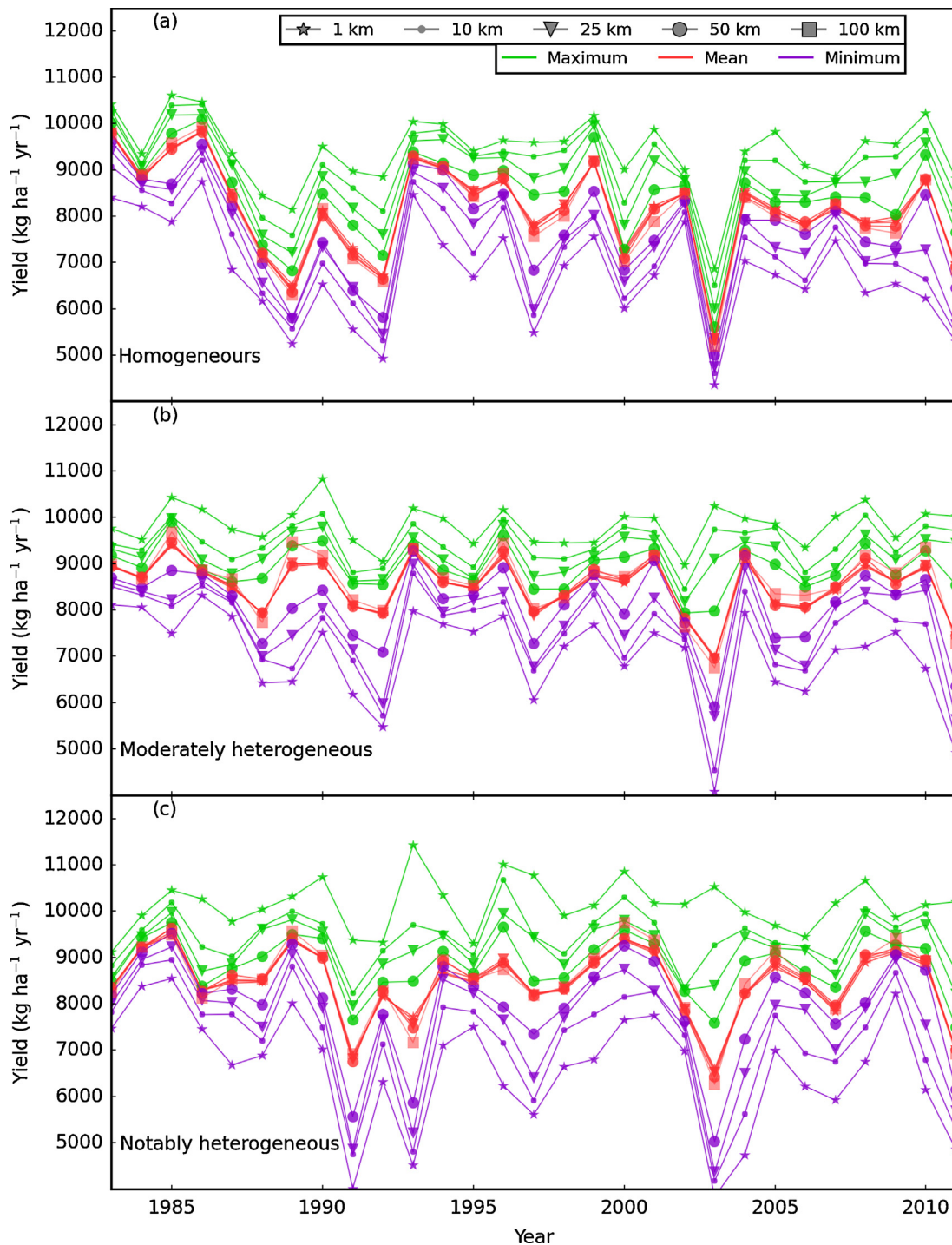


Fig. 5. Temporal variation of yields simulated using weather data for five different resolutions. Each panel is corresponding to one region in Fig. 1 with conditions of homogeneous (a), moderately heterogeneous (b), and notably heterogeneous (c) environmental conditions, respectively. For the 100 km resolution, the maximum and minimum yields were the same line, since there is only one grid cell.

3.2. Effect of changing spatial resolution on spatial patterns of simulated yield

The spatial patterns of the simulated yields were consistent across different resolutions (Fig. 6). For all resolutions, yields were low in northeastern Germany, high in northwestern and southern of Germany. With the coarsening of resolution, the frequency of the extreme low yields declined, especially along the border of the northeastern regions. Yield frequency distributions showed that

simulations with weather data at coarse resolutions simulated similar yields with weather data at fine resolution. The median of yield for the 1 km resolution was around 8750 kg ha^{-1} and it increased to 8800 kg ha^{-1} for the 50 km resolution. But the ranges of the simulated yield decreased with the coarse resolutions.

The yield bias, ΔY_i , were small in the central and northern regions, but large in the southern regions, where the largest bias attained more (less) than $\pm 1000 \text{ kg ha}^{-1}$ (Fig. 7). In the central and southern regions, the bias increased with the coarsening of the

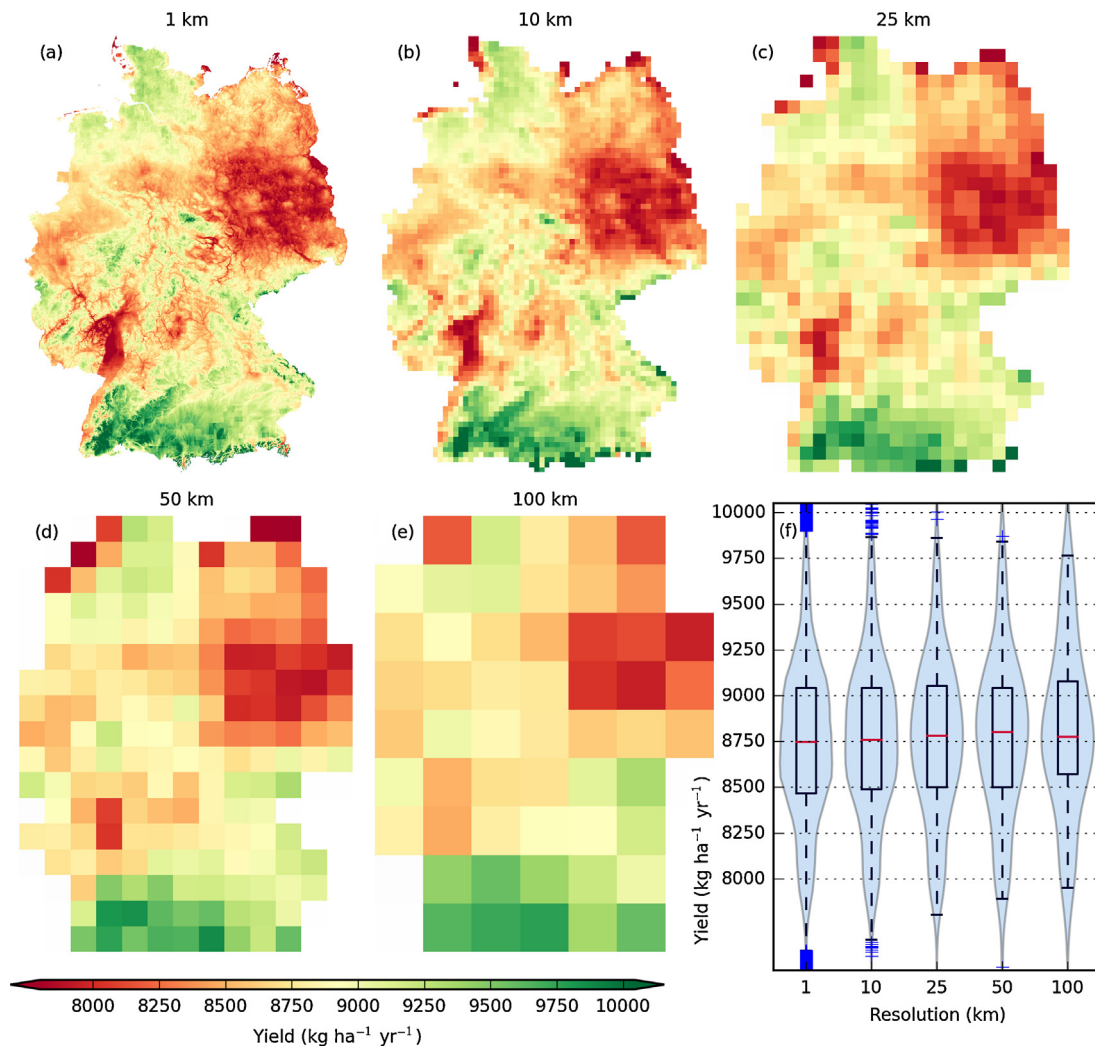


Fig. 6. Simulated yields at resolutions of 1 km (a), 10 km (b), 25 km (c), 50 km (d) and 100 km (e) and their frequency distributions (f). The red line in the box plot is the median, the edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to $1.5 \times (Q3 - Q1)$.

resolution. The bias showed discontinuous spatial patterns around the border of pixel cells of coarse resolutions (e.g. 100 km). The frequency of the yield bias also showed that similar yield medians were simulated with weather data at coarse resolutions compared to using weather data at the fine resolution. The ranges of yield bias increased with the coarsening of the resolution and weather data aggregation. For each pixel, the negative and positive values were always coexisting.

3.3. Determination of required spatial resolution

Consistent with the visual display, the empirical semivariogram model quantified the homogenous, moderately heterogeneous, and notably heterogeneous conditions for both the terrain and weather (Fig. 8). The orders of magnitude of the empirical semivariogram varied, but the trends and orders were in line with each other.

Spatial distributions of intra-pixel heterogeneity (*sill* of terrain) and yield bias (ΔY_i) were similar across all the aggregation levels, 10 km, 25 km, 50 km, 100 km (Figs. 7 and 9). For all the resolutions, they showed high values along the southern border of Germany, and in the center of the country, while intra-pixel heterogeneity was low in the majority of the pixels in the northern regions.

The high intra-pixel heterogeneity in elevation corresponded well to the yield difference (*RMSE*). The log-log linear correlations

between them were significant across all the resolutions, with R^2 equal to 0.49, 0.53, 0.41 and 0.42, for 10 km, 25 km, 50 km and 100 km, respectively (Fig. 10).

To constrain the intra-pixel *RMSE* ($\overline{RMSE}_{intra}$) to 200 kg ha⁻¹ yr⁻¹, the multi-resolution layer 1 (MR-1) showed that northern homogeneous region needed relatively low spatial resolutions ranging from 25 km to 50 km, while the major areas of the southern region needed a weather data resolution of 1 km (Fig. 11a and c). With the upper limit of $\overline{RMSE}_{intra}$ of 400 kg ha⁻¹ yr⁻¹, the northern region needed 100 km resolution, while the spatial resolution needed by the southern regions varied significantly corresponding to the local heterogeneity (Fig. 11b and d). The spatial distributions of multi-resolutions constructed using the threshold of *sill* and the true $\overline{RMSE}_{intra}$ matched well.

4. Discussion

4.1. Model performance

The aim of this study was to evaluate the effect of weather data aggregation on crop yields simulated by a dynamic crop growth model. Specific emphasis was on the effects of aggregation on the spatial variability of simulated yields. Although we did not aim for a model evaluation of regional yield simulations, some

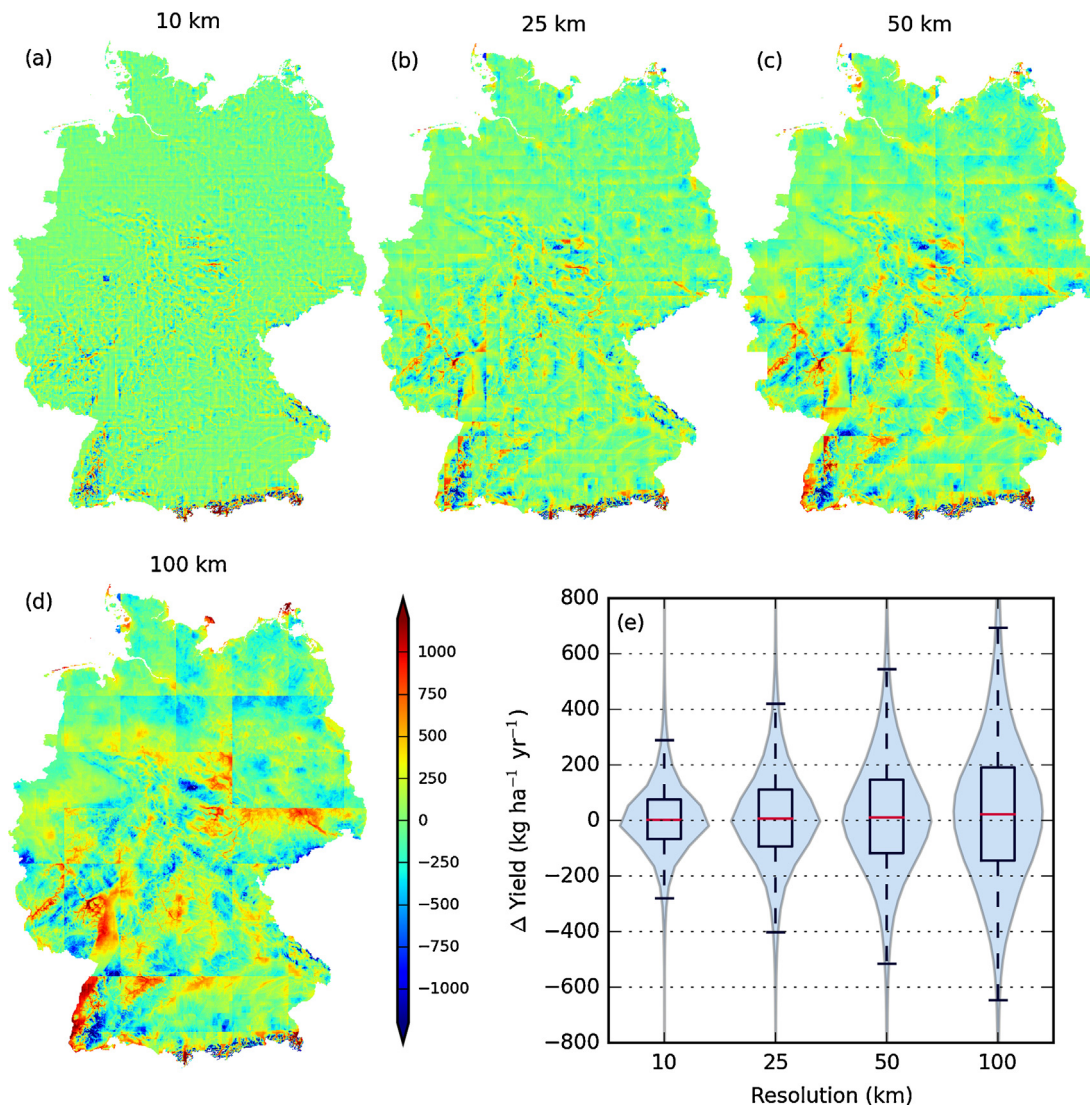


Fig. 7. Bias of yields simulated by using weather data at resolutions of 10 km (a), 25 km (b), 50 km (c), and 100 km (d) (all resampled to 1 km) and yield simulated by using weather data at a resolution of 1 km. The frequency distributions of the yield bias (e). The color bar of the maps shares the same label and unit with the violin plot. The red line in the box plot is the median, the edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to $1.5 \times (Q3 - Q1)$.

demonstration of model performance for regional application was required to justify its use in the present study. The considered model LINTUL2 has been successfully applied in many earlier studies across a range of climatic conditions (Angulo et al., 2013a; Asseng et al., 2013; Palosuo et al., 2011; Rötter et al., 2012). In line with these studies our evaluation with statistics of actual observed yields at district level showed that the model properly simulated the response of crop yields to the inter-annual variability in weather conditions for the majority of the study area (Fig. 4d and e). Model agreement with observed yields was particularly high for the north-eastern regions with relatively low yields due to drought stress (Fig. 4b and c). Importantly, at 1 km resolution, the model reproduced well the inter-annual variability over the period of 13 years (Fig. 4h).

The model overestimated yields for the majority of the study area (Fig. 4b and c). The simulated water-limited yields were on average about 20% higher than the observed yields (Fig. 4h) which is in good agreement with the estimated yield gap between water-limited and actual yield for Germany (Boogaard et al., 2013). Moreover, the overestimation spatially varied and was

especially high in the southern regions of Germany (Fig. 4b and c). LINTUL2 does not account for several factors affecting actual yields such as frost damage, nutrient deficiencies, pests and diseases. As the importance of these factors in affecting actual yields may vary depending on the region this may explain some of the spatial differences in the overestimation of yields. Also, the present study was conducted under water-limited production conditions with a single soil profile of good water holding capacity. The water stress varied with the growing season precipitation, potential evapotranspiration and crop water demand. Spatial variation of soil physical properties were ignored in this study but can be expected to further affect spatial heterogeneity. The same applies to nutrient limited production conditions which we also did not consider, when assuming ample supply of nutrients.

4.2. Yield variability at different spatial aggregations

Spatially aggregated data have been widely used to drive crop models at regional scales (Ewert et al., 2011; Fiorese and Guariso,

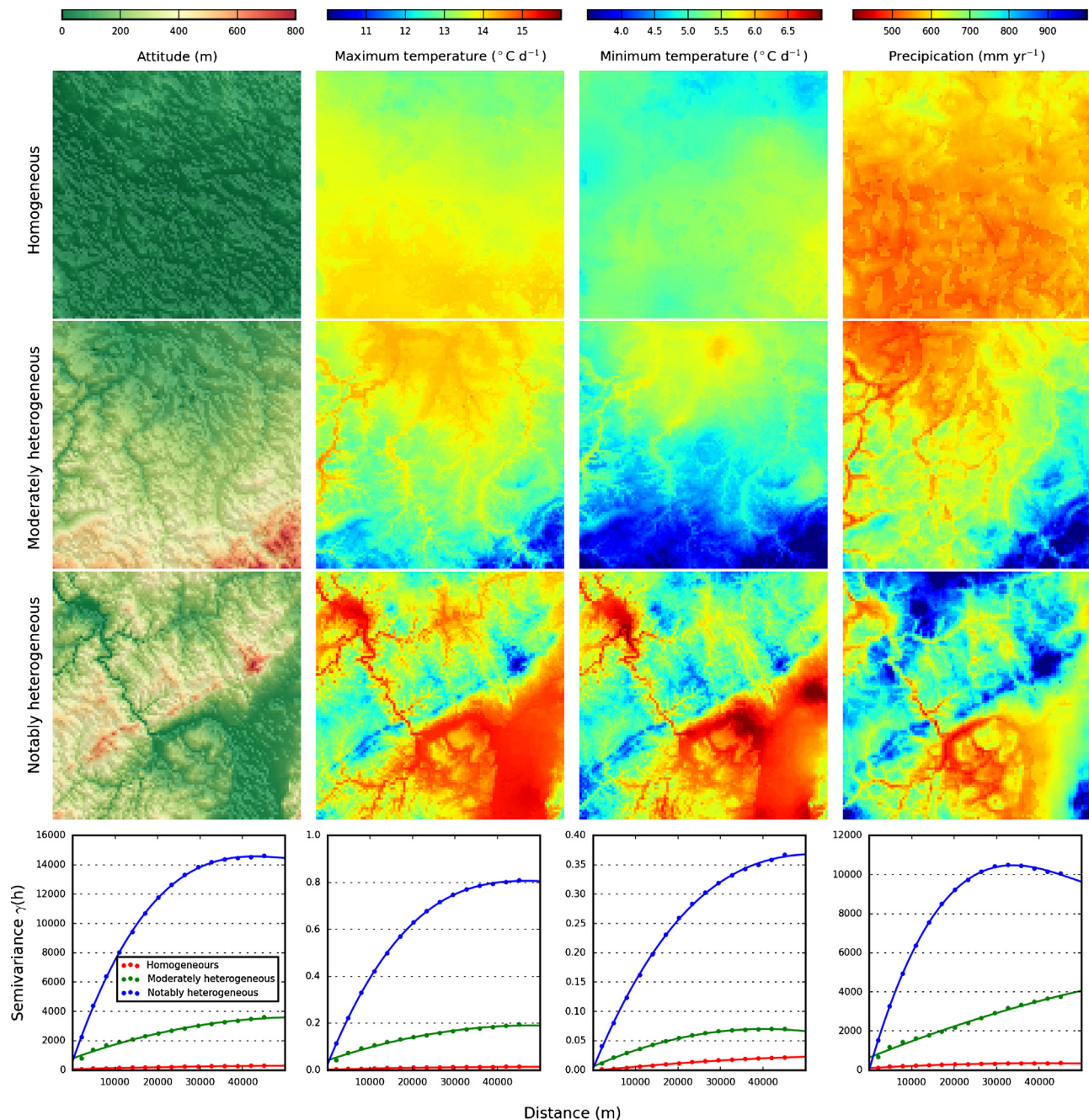


Fig. 8. Three regions with a spatial extent of $10,000 \text{ km}^2$ that represent homogenous (the first row), moderately heterogeneous (second row), and notably heterogeneous (the third row) spatial conditions of terrain, and of maximum temperature, minimum temperature, and annual precipitation. The pixel size in the images is 1 km. Each image included 10,000 pixels. The fourth row is the fitted semivariogram for the three images above the plots. The location of the three regions can be found in Fig. 2.

2010; Liu, 2009), while the aggregation effects on the simulated results are still inconclusive (Angulo et al., 2013b; Wit et al., 2005). For example, Hansen and Jones (2000) and Di Vittorio and Miller (2014) pointed out that the aggregation of the weather data introduced high uncertainty, while Angulo et al. (2013b) found the aggregation effects were ignorable compared to the use of different models. Within a selected region (100 km), for a specific year, the mean of simulated yields made little difference across different spatial resolutions, regardless of the spatial heterogeneity of the regions. This is due to the aggregation effects on the weather which were either positive or negative. When using these low resolution data to drive the model, the caused effects on simulated yields were also either positive or negative (Fig. 7). When taking

the mean of the simulated yields, the negative and positive impacts compensated each other, thus caused the similar means across different resolutions. For some cases, the simulated mean of yield showed some moderate difference across different resolutions. Such response is the result of the non-linear relationships used in the crop model to simulate weather effects on yield. The non-linear effect was well explained in Hansen and Jones (2000). However, in the present study this effect was small as also reported from other regions (Angulo et al., 2013b). Also, the similar mean can result in a consistent spatial pattern across resolutions (van Bussel et al., 2011; Wit et al., 2005). Even if the aggregation of the weather data does not change the major properties of the weather data, it can moderate the extreme weather conditions such as extremely high

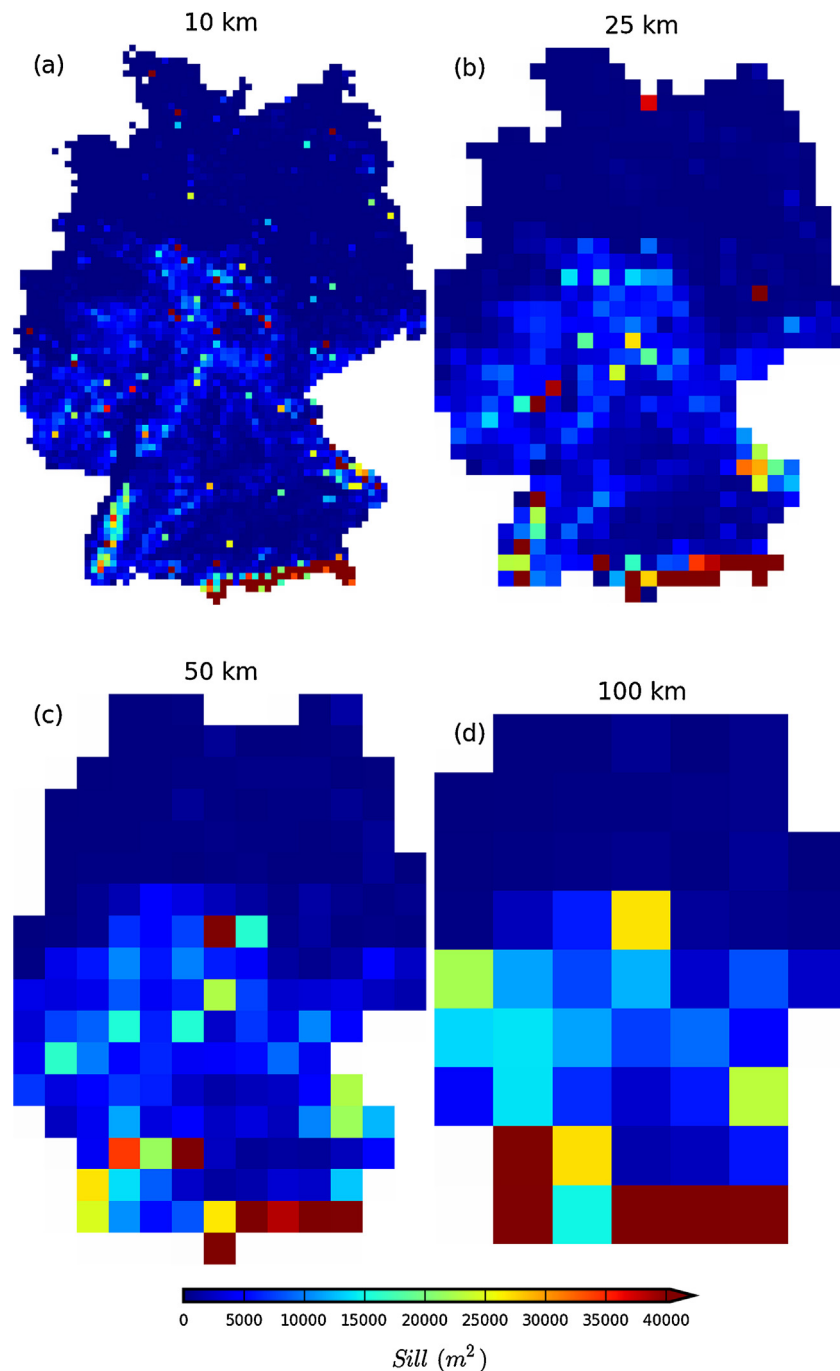


Fig. 9. Intra-pixel spatial heterogeneity of elevation at resolutions of 10 km (a), 25 km (b), 50 km (c) and 100 km (d). The spatial heterogeneity was quantified by the *sill* values of four fitted semivariogram models. The *sill* value shown in the maps is the mean of the fitted four models.

temperature or rainfall (Priya and Shibasaki, 2001). In the present study, the reduction in extreme weather conditions resulted in less extreme (high or low) yields (Fig. 6). This could explain that the magnitude of spatial yield variability decreased with the coarsening of spatial resolution (Fig. 5).

The temporal variations of the mean yield at different resolutions (1 to 100 km) were consistent and close to each other (Fig. 5) regardless of the environmental heterogeneity, i.e. homogeneous, moderately heterogeneous, or significant heterogeneous. This confirmed that a low resolution weather maybe enough for simulation of the inter-annual variability of yield for a large area (Olesen et al., 2000).

4.3. Intra-pixel spatial heterogeneity of terrain and weather

Most studies on aggregation effects used the indicators of mean, median or variation range for analyzing simulated yields at different spatial resolutions. This study represents advancement by specifically analyzing the effect of data aggregation on spatial heterogeneity of yields. Importantly, we found that the main effects of aggregation on the input data were the reduction of the intra-pixel spatial heterogeneity for the daily weather data.

The close relationship between altitude and local weather condition is widely known (Ashcroft et al., 2012; Dobrowski, 2011) and well demonstrated in Fig. 8. In this study, we do not aim to

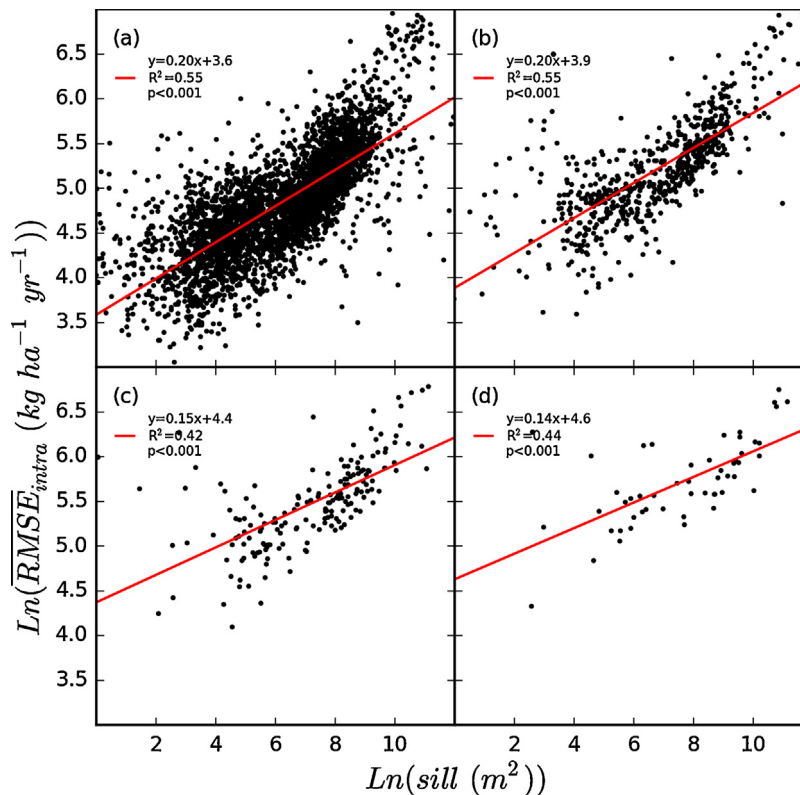


Fig. 10. The relationship between the intra-pixel spatial heterogeneity (*sill*) of elevation and intra-pixel root mean square error ($\overline{RMSE}_{intra}$) of yield between 1 km resolution and coarse resolutions of 10 km (a), 25 km (b), 50 km (c) and 100 km (d). The $\ln$ in the figure indicates the natural logarithm with base e (equal to 2.718281828).

proof this relationship but rather to take advantage of it to use a DEM as a surrogate to determine the required spatial resolution for large scale simulations. Gridded weather data for large scale simulation are normally produced by interpolation of measurements from weather stations. As part of this interpolation, the altitude differences between the intermediate locations and the weather stations are normally used to adjust the temperature values. However the temperature of an intermediate location is also influenced by other factors like vegetation, slope, aspect, or soil. In addition, the prevailing effects of the altitude would only be obvious within a relatively small spatial extent. For a large spatial extent, the effects could diminish, since latitude and longitude (distance to the sea) would become more important (Graae et al., 2012). Using the method proposed in this study, noise could be added to the predicted multi-resolution surface in locations where these influences are strong. However, the good performance of using the DEM as an indicator in Fig. 11 suggests that the noise can be neglected for the region studied here.

4.4. Intra-pixel spatial heterogeneity for yield

The biases of annual mean yields due to the weather data aggregation were clearly shown in the maps (Fig. 7). The bias in the southern regions with a complex topography was higher than the bias in the northern regions with a flat topography. This finding motivated us to further investigate the correlation between the spatial heterogeneity of terrain and the yield bias. Aggregating the climate data through a spatial averaging method weakened the spatial heterogeneity, thus caused the substantial yield bias simulated with weather data at coarse resolutions. We suggest that defective integration of heterogeneous inputs including distortions of spatial mean values of predictions and time-series variability of

the spatial means, led to the aggregation bias (Hansen and Jones, 2000; Mearns et al., 1999). The coarser the resolution, the more spatial information was lost in the data aggregation processes (Fortin and Dale, 2005). This explained the expanding yield biases when weather data resolutions were getting coarser. For future climate change impact studies, the results from this paper confirmed the necessity of downscaling the general circulation model outputs to get the regional details using the regional climate models (RCMS) or other techniques, especially for the regions with high spatial heterogeneity (Corney et al., 2013).

Both the good spatial match of yield bias and intra-pixel spatial heterogeneity of input data and the significant log–log linear relationship indicate that the intra-pixel spatial heterogeneity of input data determined the effects of changing input data resolution. In homogeneous areas, coarsening the input data resolution caused little heterogeneity loss, thus a small bias of yield was simulated, and vice versa. This suggests that the spatial heterogeneity of the input data is a key factor that influences the data aggregation effects. Aggregation of pixels belonging to homogeneous areas would cause small biases in the simulated results. The crop model, LINTUL2, used in this study is a relatively simple crop model. It does not simulate effects of nutrient supply and other yield-limiting and reducing factors. Involving these factors could alter the way how yield bias responds to the data aggregation (Angulo et al., 2013b).

4.5. Determining spatial resolution for regional yield simulation

The good match between the two layers constructed using the threshold of *sill* and the true $\overline{RMSE}_{intra}$ indicates that we can use the local spatial heterogeneity of terrain to predict the required spatial resolution. Fig. 11 showed that the resolution could be determined

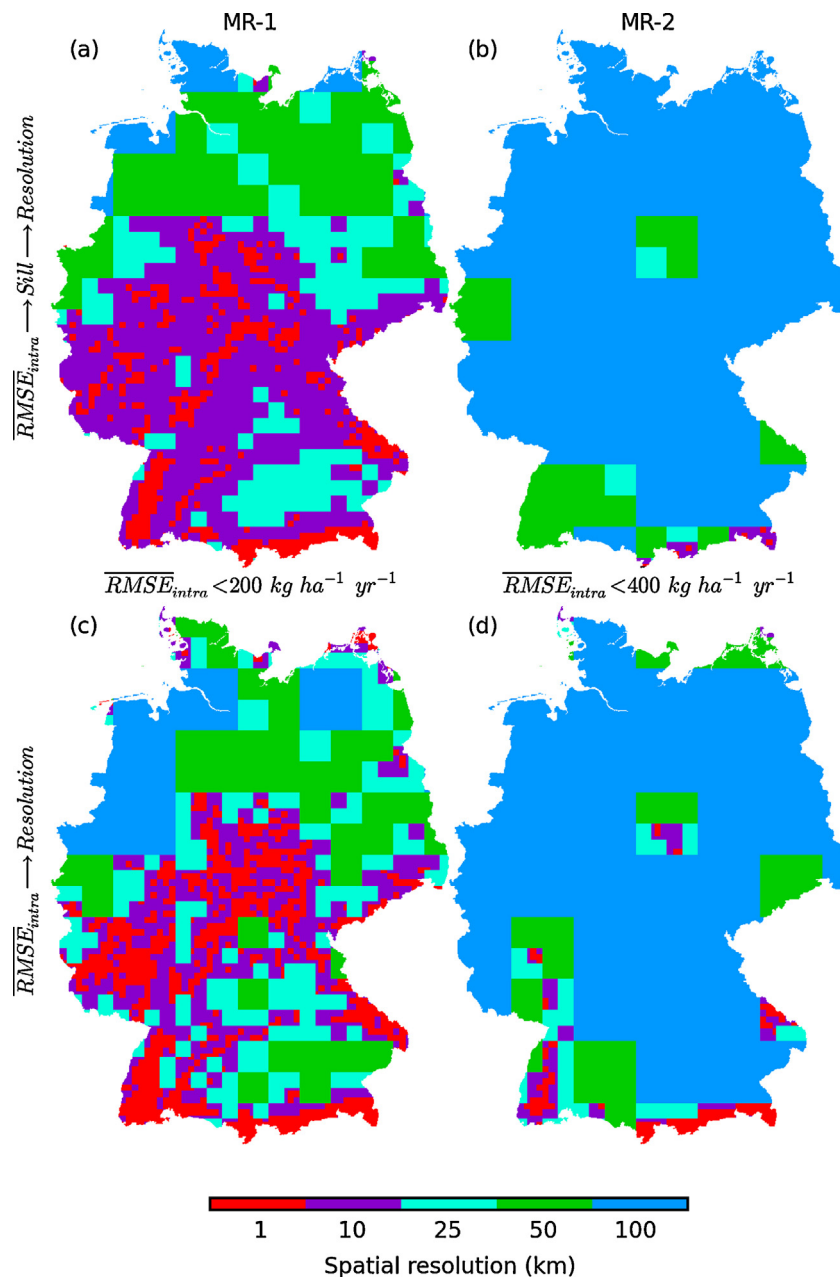


Fig. 11. The spatial distribution of required spatial resolution to constrain the intra-pixel root mean squared error ($\overline{RMSE}_{intra}$) of yield to specific upper thresholds (200 and 400 kg ha⁻¹ yr⁻¹). The first row (a and b) was calculated according to the regressed equations between sill and $\overline{RMSE}_{intra}$ shown in Fig. 10. The second row (c and d) was calculated according to Eq. (6). The first column of which the $\overline{RMSE}_{intra}$ threshold is 200 kg ha⁻¹ yr⁻¹ was named as multi-resolution 1 (MR-1). The second column of which the $\overline{RMSE}_{intra}$ threshold is 400 kg ha⁻¹ yr⁻¹ was named as multi-resolution 2 (MR-2).

according to the local environmental heterogeneity and specified thresholds of intra-pixel RMSE ($\overline{RMSE}_{intra}$). The regressed equations in this study may change in regions other than Germany or using a different crop model (Mearns et al., 1999), but the developed method may still be appropriate to identify the required spatial resolution for a specified threshold of acceptable yield difference due to aggregation. Using different spatial resolutions for different regions could control the intra-pixel bias, reduce the model runs and save the computing power (Fig. 12). The high $\overline{RMSE}_{intra}$ in the 10 km resolution was avoided in the MR-1 by using 1 km resolution (Fig. 12, ellipse), while the low $\overline{RMSE}_{intra}$ area were replaced by coarser resolutions without increasing the $\overline{RMSE}_{intra}$ much and saved the number of simulations (Fig. 12, rectangle). This optimization may be important in particular for studies where many model runs are required, e.g. uncertainty and

sensitivity analysis or stochastic crop modeling (Di Vittorio and Miller, 2014).

Spatial zones or polygons could be another efficient and flexible way than regular pixels in capturing differences in the spatial variance. Spatial zones or polygons aggregate the areas with similar environmental properties that influence crop growth. Aggregating similar environmental properties inside a polygon would not result in significantly loss of spatial details. It could balance between the number of simulation runs and the details of environmental characteristics (Ewert et al., 2011). For example, Zhao et al. (2013b) created climate-soil zones to simulate the long-term dynamics of soil organic carbon at the national scale of Australia. van Wart et al. (2013) used the agro-climatic zones to simulate the yield gap, difference between potential and water limited yield, at the global scale.

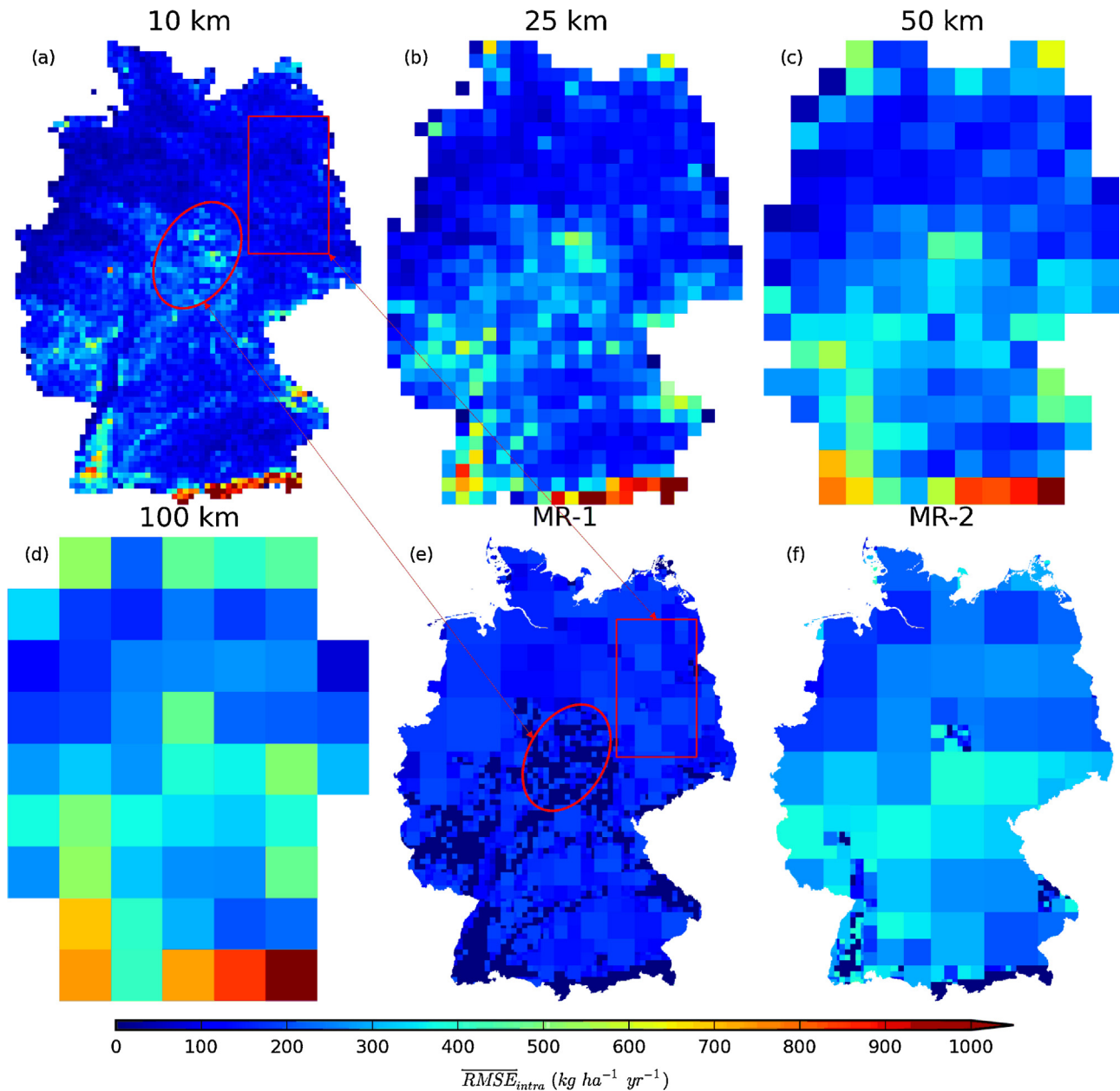


Fig. 12. The spatial distribution of intra-pixel root mean square error ($\overline{RMSE}_{intra}$) of yield at resolutions of 10 km (a), 25 km (b), 50 km (c) and 100 km, and multi-resolution 1 (MR-1, e) and multi-resolution 2 (MR-2, f). The resolutions in MR-1 and MR-2 were from Fig. 11c and d, respectively. The ellipse showed that by fining the resolution in the area could significantly decrease the $\overline{RMSE}_{intra}$, while the rectangle showed that coarsening the resolution in this area does not increase the $\overline{RMSE}_{intra}$.

5. Conclusion

Regional scale application of crop models developed at field scale is becoming a common practice in impact assessment studies of climate variability and change. This paper evaluated the impact of weather data aggregation on simulation of rainfed crop yields at the national scale of Germany. It advances from early studies on aggregation by specifically addressing the effect of aggregation on spatial yield variability. Different resolutions of weather data simulated similar spatial patterns of annual mean wheat yields, but the ranges of the simulated yields were different depending on the regions. The aggregation effects on yields were high in the southern regions with complex topographic conditions, but low in the northern regions with flat topographic conditions. They were highly correlated with the intra-pixel spatial heterogeneity of local terrain, indicated by the *sill* value of the semivariogram model. We

demonstrated how to determine the spatial resolution by using the local spatial heterogeneity and specified thresholds for the yield difference due to aggregation. We conclude that the data aggregation effect was inversely proportional to the representativeness of aggregated weather input data to the original data. When using aggregated input data, the spatial heterogeneity of the input data needs to be fully considered. For a region with significant spatial heterogeneity in weather data, high resolution should be used. For future climate change impact studies at regional scale, generating the regional climate data from general circulation model is therefore essential for regions with high spatial heterogeneity. The semivariogram model is a useful method in quantification of the spatial heterogeneity that could guide decision for choosing the spatial resolution of the input data. The study was constrained to one crop model, one crop, one output variable and for rainfed conditions assuming all other production factors optimal. Hence,

our results require substantiation for other production situations, crops, output variables and for different crop models.

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Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.agrformet.2014.09.026>.

References

- Adam, M., et al., 2012. Building crop models within different crop modelling frameworks. *Agric. Syst.* 113 (0), 57–63.
- Adam, M., Van Bussel, L.G.J., Leffelaar, P.A., Van Keulen, H., Ewert, F., 2011. Effects of modelling detail on simulated potential crop yields under a wide range of climatic conditions. *Ecol. Modell.* 222 (1), 131–143.
- Angulo, C., et al., 2013a. Implication of crop model calibration strategies for assessing regional impacts of climate change in Europe. *Agric. For. Meteorol.* 170 (0), 32–46.
- Angulo, C., et al., 2013b. Characteristic ‘fingerprints’ of crop model responses to weather input data at different spatial resolutions. *Eur. J. Agron.* 49 (0), 104–114.
- Ashcroft, M.B., Gollan, J.R., 2012. Fine-resolution (25 m) topoclimatic grids of near-surface (5 cm) extreme temperatures and humidities across various habitats in a large (200 × 300 km) and diverse region. *Int. J. Climatol.* 32 (14), 2134–2148.
- Ashcroft, M.B., Gollan, J.R., Warton, D.I., Ramp, D., 2012. A novel approach to quantify and locate potential microrefugia using topoclimate, climate stability, and isolation from the matrix. *Global Change Biol.* 18 (6), 1866–1879.
- Asseng, S., et al., 2013. Uncertainty in simulating wheat yields under climate change. *Nat. Clim. Change* 3 (9), 827–832.
- Balković, J., et al., 2013. Pan-European crop modelling with EPIC: implementation, up-scaling and regional crop yield validation. *Agric. Syst.* 120 (0), 61–75.
- Bian, L., Butler, R., 1999. Comparing effects of aggregation methods on statistical and spatial properties of simulated spatial data. *Photogramm. Eng. Remote Sens.* 65, 73–84.
- Boogaard, H., Wolf, J., Supit, I., Niemeyer, S., van Ittersum, M., 2013. A regional implementation of WOFOST for calculating yield gaps of autumn-sown wheat across the European Union. *Field Crops Res.* 143 (0), 130–142.
- Corney, S., et al., 2013. Performance of downscaled regional climate simulations using a variable-resolution regional climate model: tasmania as a test case. *J. Geophys. Res. Atmos.* 118 (21), 11936–12111.
- Curran, P.J., 1988. The semivariogram in remote sensing: an introduction. *Remote Sens. Environ.* 24 (3), 493–507.
- De Vries, F.P., Rabbinge, R., Groot, J., 1997. Potential and attainable food production and food security in different regions. *Philos. Trans. R. Soc. London, Ser. B: Biol. Sci.* 352 (1356), 917–928.
- Di Vittorio, A.V., Miller, N.L., 2014. Reducing the impact of model scale on simulated, gridded switchgrass yields. *Environ. Modell. Softw.* 51 (0), 70–83.
- Dobrowski, S.Z., 2011. A climatic basis for microrefugia: the influence of terrain on climate. *Global Change Biol.* 17 (2), 1022–1035.
- DWD, 2012. REGNIE Gridded Data of Daily Precipitation. German Meteorological Service (DWD), <http://werdis.dwd.de/werdis/retrieve.data.do?pidpat=de.dwd.hydromet.regnie.daily&tolevel=true> (accessed 08/02/2014).
- Engelen, V., Wen, v.T., 1993. Global and National Soils and Terrain Digital Databases (SOTER): Procedures Manual. International Soil Reference and Information Centre.
- Ewert, F., et al., 2011. Scale changes and model linking methods for integrated assessment of agri-environmental systems. *Agricult. Ecosys. Environ.* 142 (1–2), 6–17.
- Ewert, F., van Oijen, M., Porter, J.R., 1999. Simulation of growth and development processes of spring wheat in response to CO₂ and ozone for different sites and years in Europe using mechanistic crop simulation models. *Eur. J. Agron.* 10 (3–4), 231–247.
- Farré, I., Van Oijen, M., Leffelaar, P., Faci, J., 2000. Analysis of maize growth for different irrigation strategies in northeastern Spain. *Eur. J. Agron.* 12 (3), 225–238.
- Fiorese, G., Guariso, G., 2010. A GIS-based approach to evaluate biomass potential from energy crops at regional scale. *Environ. Modell. Softw.* 25 (6), 702–711.
- Fischer, G., Shah, M., Tubiello, F.N., van Velhuizen, H., 2005. Socio-economic and climate change impacts on agriculture: an integrated assessment, 1990–2080. *Philos. Trans. R. Soc. London, B: Biol. Sci.* 360 (1463), 2067–2083.
- Folberth, C., Yang, H., Wang, X., Abbaspour, K.C., 2012. Impact of input data resolution and extent of harvested areas on crop yield estimates in large-scale agricultural modeling for maize in the USA. *Ecol. Modell.* 235, 8–18.
- Fortin, M.-J., Dale, M.R.M.T., 2005. *Spatial Analysis: A Guide for Ecologists*. Cambridge University Press, Cambridge.
- Gaiser, T., et al., 2013. Modeling biopore effects on root growth and biomass production on soils with pronounced sub-soil clay accumulation. *Ecol. Modell.* 256, 6–15.
- Graae, B.J., et al., 2012. On the use of weather data in ecological studies along altitudinal and latitudinal gradients. *Oikos* 121 (1), 3–19.
- Guisan, A., Graham, C.H., Elith, J., Huettmann, F., the, N.S.D.M.G., 2007. Sensitivity of predictive species distribution models to change in grain size. *Diversity and Distributions* 13 (3), 332–340.
- Hancock, P., Hutchinson, M., 2006. Spatial interpolation of large climate data sets using bivariate thin plate smoothing splines. *Environ. Modell. Softw.* 21 (12), 1684–1694.
- Hansen, J.W., Jones, J.W., 2000. Scaling-up crop models for climate variability applications. *Agric. Syst.* 65 (1), 43–72.
- Harrison, P.A., Porter, J.R., Downing, T.E., 2000. Scaling-up the AFRCWHEAT2 model to assess phenological development for wheat in Europe. *Agric. For. Meteorol.* 101 (2), 167–186.
- Hijmans, R.J., Condon, B., Carrillo, R., Kropff, M., 2003. A quantitative and constraint-specific method to assess the potential impact of new agricultural technology: the case of frost resistant potato for the Altiplano (Peru and Bolivia). *Agric. Syst.* 76 (3), 895–911.
- Kerr, A., 1993. Topography, climate and ice masses: a review. *Terra Nova* 5 (4), 332–342.
- Keulen, H., 1975. Simulation of Water Use and Herbage Growth in Arid Regions. Pudoc, Wageningen, pp. 184.
- LfULG, 2000–2012. Landessortenversuche Winterweizen. Sächsisches Landesamt für Umwelt, Landwirtschaft und Geologie (LfULG), Dresden, Germany.
- Li, H., Reynolds, J., 1995. On definition and quantification of heterogeneity. *Oikos*, 280–284.
- Liu, J., 2009. A GIS-based tool for modelling large-scale crop-water relations. *Environ. Modell. Softw.* 24 (3), 411–422.
- Mearns, L., Easterling, W., Hays, C., Marx, D., 2001. Comparison of agricultural impacts of climate change calculated from high and low resolution climate change scenarios: Part I. The uncertainty due to spatial scale. *Clim. Change* 51 (2), 131–172.
- Mearns, L.O., Mavromatis, T., Tsvetinskaya, E., Hays, C., Easterling, W., 1999. Comparative responses of EPIC and CERES crop models to high and low spatial resolution climate change scenarios. *J. Geophys. Res. Atmos.* 104 (D6), 6623–6646.
- Mummery, D., Battaglia, M., 2002. Data input quality and resolution effects on regional and local scale *Eucalyptus globulus* productivity predictions in north-east Tasmania. *Ecol. Modell.* 156 (1), 13–25.
- Nendel, C., et al., 2013. Simulating regional winter wheat yields using input data of different spatial resolution. *Field Crops Res.* 145, 67–77.
- Olesen, J.E., Böcher, P.K., Jensen, T., 2000. Comparison of scales of climate and soil data for aggregating simulated yields of winter wheat in Denmark. *Agricult. Ecosys. Environ.* 82 (1–3), 213–228.
- Orcutt, G.H., Watts, H.W., Edwards, J.B., 1968. Data aggregation and information loss. *Am. Econ. Rev.* 58 (4), 773–787.
- Palosuo, T., et al., 2011. Simulation of winter wheat yield and its variability in different climates of Europe: a comparison of eight crop growth models. *Eur. J. Agron.* 35 (3), 103–114.
- Pebesma, E.J., 2004. Multivariable geostatistics in S: the gstat package. *Comput. Geosci.* 30 (7), 683–691.
- Penman, H.L., 1963. Vegetation and hydrology. *Soil Sci.* 96 (5), 357.
- Posselt, R., Müller, R., Stöckli, R., Trentmann, J., 2011. CM SAF surface radiation MVIRI data set 1.0—monthly means/daily means/hourly means. *Satell. Appl. Facility Clim. Monit.*, <http://dx.doi.org/10.5676/EUM.SAF.CM/RAD.MVIRI/V001>.
- Priya, S., Shibasaki, R., 2001. National spatial crop yield simulation using GIS-based crop production model. *Ecol. Modell.* 136 (2–3), 113–129.
- Rabbinge, R., Van Diepen, C., 2000. Changes in agriculture and land use in Europe. *Eur. J. Agron.* 13 (2), 85–99.
- Rötter, R.P., et al., 2012. Simulation of spring barley yield in different climatic zones of Northern and Central Europe: a comparison of nine crop models. *Field Crops Res.* 133 (0), 23–36.
- Seoane, J., Bustamante, J., Diaz-Delgado, R., 2004. Competing roles for landscape, vegetation, topography and climate in predictive models of bird distribution. *Ecol. Modell.* 171 (3), 209–222.
- Siebert, S., Ewert, F., 2012. Spatio-temporal patterns of phenological development in Germany in relation to temperature and day length. *Agric. For. Meteorol.* 152 (0), 44–57.
- Spitters, C., 1987. An analysis of variation in yield among potato cultivars in terms of light absorption, light utilization and dry matter partitioning. *Agrometeorology Potato Crop* 214, 71–84.
- Spitters, C., 1989. Crop growth models: their usefulness and limitations. In: VI Symposium on the Timing of Field Production of Vegetables, pp. 349–368, 267.
- Statistische Ämter des Bundes und der Länder, 2013. Regionaldatenbank Deutschland. Statistische Ämter des Bundes und der Länder, <https://www.regionalstatistik.de/genesis/online/logon> (accessed 08/05/2014).

- Stein, M.L., 1999. *Interpolation of Spatial Data: Some Theory for Kriging*. Springer, New York, NY.
- van Bussel, L.G.J., Ewert, F., Leffelaar, P.A., 2011. Effects of data aggregation on simulations of crop phenology. *Agricult. Ecosys. Environ.* 142 (1–2), 75–84.
- van Ittersum, M.K., et al., 2008. Integrated assessment of agricultural systems—a component-based framework for the European Union (SEAMLESS). *Agric. Syst.* 96 (1–3), 150–165.
- van Ittersum, M.K., et al., 2003. On approaches and applications of the Wageningen crop models. *Eur. J. Agron.* 18 (3), 201–234.
- van Oijen, M., Lefelaar, P., 2008. *Lintul-2: Water Limited Crop Growth: A Simple General Crop Growth Model for Water-Limited Growing Conditions*. Wageningen University, Wageningen (Plant Sciences).
- van Wart, J., et al., 2013. Use of agro-climatic zones to upscale simulated crop yield potential. *Field Crops Res.* 143 (0), 44–55.
- Wit, A.d., Boogaard, H., Diepen, C.v., 2005. Spatial resolution of precipitation and radiation: the effect on regional crop yield forecasts. *Agric. For. Meteorol.* 135 (1), 156–168.
- Wolf, J., Van Oijen, M., Kempenaar, C., 2002. Analysis of the experimental variability in wheat responses to elevated CO₂ and temperature. *Agricult. Ecosyst. Environ.* 93 (1), 227–247.
- Wu, J., 2004. Effects of changing scale on landscape pattern analysis: scaling relations. *Landscape Ecol.* 19 (2), 125–138.
- Zhao, G., et al., 2013a. Large-scale, high-resolution agricultural systems modeling using a hybrid approach combining grid computing and parallel processing. *Environ. Modell. Softw.* 41 (0), 231–238.
- Zhao, G., et al., 2013b. Impact of agricultural management practices on soil organic carbon: simulation of Australian wheat systems. *Global Change Biol.* 19 (5), 1585–1597.